Consensus Circuit Scoring: Cross-Method Binary Consensus Identifies Faithful Circuit Subsets When Discovery Methods Disagree

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Abstract

Mechanistic interpretability relies on circuit discovery methods to identify functionally important subgraphs in neural networks, yet different methods—gradient-based attribution (EAP-IG), edge pruning (EP), and activation patching (ACDC)—often identify substantially different circuits for the same task. We propose *Consensus Circuit Scoring* (CSS), a method-agnostic binary consensus framework that scores each edge by the fraction of $K \geq 3$ discovery methods that independently select it. We show that full-consensus edges (CSS=1.0) form dramatically smaller yet comparably or more faithful circuits, but *only* when the constituent methods genuinely disagree. Across 4 tasks, 4 model architectures (GPT-2 Small, Qwen-2.5 0.5B, Gemma-2 2B, Llama-3.1 8B), and 10 experimental conditions, we find a strong negative correlation between maximum pairwise Jaccard similarity of method outputs and CSS utility (Spearman $\rho = 0.943$, $p = 0.005$), with a fitted threshold at Jaccard ≈ 0.55. Below this threshold, CSS=1.0 consistently outperforms the best single method in circuit faithfulness while using $4{-}10\times$ fewer edges. We further show that CSS outperforms within-method threshold diversity (CIRCUS) on all tested tasks, and that the advantage is robust across ablation evaluation methods and binarization thresholds. Our results establish inter-method correlation as a diagnostic signal for when consensus-based circuit identification is beneficial.

1 Introduction

Circuit discovery in neural networks aims to identify the minimal subgraph—a *circuit*—that is responsible for a model's behavior on a given task [**??**]. Multiple algorithmic approaches have been developed for this purpose, including gradient-based edge attribution [**?**], iterative edge pruning [**?**], activation patching [**?**], and representation similarity methods [**?**]. A persistent challenge is that these methods often disagree substantially on which edges constitute the circuit, even when applied to the same model and task [**?**].

This disagreement raises a fundamental question: *when multiple methods identify different circuits, which edges are truly functionally important?* One natural hypothesis is that edges selected by multiple independent methods—*consensus edges*—are more likely to represent genuine functional components, as their identification is robust to algorithmic assumptions. Conversely, edges selected by only one method may reflect method-specific artifacts.

We formalize this intuition as *Consensus Circuit Scoring* (CSS). Given $K \geq 3$ discovery methods from distinct algorithmic families, CSS assigns each edge a score equal to the fraction of methods that select it after binarization. Full-consensus edges (CSS=1.0) are those selected by all K methods independently. The approach is deliberately simple—binary voting with no score calibration—which makes it applicable whenever multiple binarized circuit outputs are available.

Our investigation reveals two primary findings:

Claim 1: CSS=1.0 identifies edge-efficient consensus circuits. On tasks where discovery methods provide genuinely independent evidence (low pairwise Jaccard index between method outputs), CSS=1.0 circuits are $4-10\times$ smaller than single-method circuits while achieving equal or superior faithfulness, as measured by Circuit Preservation Ratio (CPR) under ablation.

Claim 2: CSS utility is predicted by inter-method correlation. Across 6 model×task combinations spanning Jaccard values from 0.24 to 0.70, the CSS faithfulness advantage correlates strongly and negatively with maximum pairwise Jaccard (Spearman $\rho = 0.943$, $p = 0.005$). A fitted threshold at Jaccard ≈ 0.55 separates conditions where CSS helps from those where it does not, providing an actionable diagnostic for practitioners.

These findings are supported by experiments on 4 tasks (Indirect Object Identification, Subject-Verb Agreement, Factual Recall, Greater-Than Arithmetic) across 4 architectures (GPT-2 Small 117M, Qwen-2.5 0.5B, Gemma-2 2B, Llama-3.1 8B), with ablation studies on binarization thresholds, evaluation methods, and comparison to the concurrent CIRCUS approach [**?**].

2 Method

2.1 Problem Setup

We consider a transformer model as a computational graph $G = (V, E)$ where vertices represent attention heads and MLP layers, and edges represent information flow between them. A *circuit discovery method* $\mathcal{M}$ takes as input a model, a task dataset, and produces a scored edge set $\{(e, s_e) : e \in E\}$, where s_e indicates the estimated importance of edge e for the task.

2.2 Binarization

Each method produces a continuous importance ranking. To enable cross-method comparison, we binarize each method's output by retaining only the top-k edges:

$$B_{\mathcal{M}}(k) = \{e \in E : \text{rank}_{\mathcal{M}}(e) \leq k\} \tag{1}$$

For models with differing total edge counts, we use a percentage-based threshold (e.g., top-0.3%) to ensure comparable circuit sizes. The choice of k affects the absolute Jaccard values but not the relative ordering of methods (Section 5.1).

2.3 Consensus Circuit Scoring

Given K binarized method outputs $B_{\mathcal{M}_1}(k), \ldots, B_{\mathcal{M}_K}(k)$, we define the *consensus score* of edge e as:

$$\text{CSS}(e) = \frac{1}{K} \sum_{i=1}^{K} \mathbf{1}[e \in B_{\mathcal{M}_i}(k)] \tag{2}$$

The consensus circuit at threshold τ is:

$$C_\tau = \{e \in E : \text{CSS}(e) \geq \tau\} \tag{3}$$

For $K = 3$, the possible CSS values are $\{1/3, 2/3, 1\}$, corresponding to edges selected by exactly one, two, or all three methods. We focus on CSS=1.0 ($\tau = 1$, full consensus) as the primary object of study.

2.4 Method Family Requirements

CSS requires methods from *distinct algorithmic families* to provide independent evidence. We use three families:

- **Gradient attribution** (EAP-IG): Integrated gradients on edge activations [**?**].
- **Edge pruning** (EP): Iterative removal of least-important edges [**?**].
- **Activation patching** (ACDC): Edge-level clean→corrupt activation substitution [**?**].

EAP-IG and EP are both gradient-adjacent but differ in their scoring mechanism (attribution vs. pruning sensitivity). ACDC uses a fundamentally different paradigm (causal intervention). This diversity is critical: when one method (IFR, a similarity-based approach) degenerates on certain architectures (Section 5.4), consensus fails entirely.

2.5 Evaluation Metrics

We evaluate circuit faithfulness using two metrics under edge ablation:

- **Circuit Preservation Ratio (CPR)**: The fraction of the original model's task performance preserved when only the circuit edges are active. CPR > 0 indicates the circuit captures functional structure; CPR $= 1$ indicates perfect preservation.
- **Circuit Minimality Distance (CMD)**: $1 - \text{CPR}$, measuring performance loss. Lower is better.

We evaluate under three ablation baselines: *zero* (ablated edges set to zero), *mean* (set to dataset-mean activation), and *resample* (set to activations from a different input). Mean ablation is preferred for localized circuits where zero ablation causes cascading destruction (Section 5.2).

2.6 Method Diversity Measurement

We quantify inter-method agreement using the pairwise Jaccard index:

$$J(\mathcal{M}_i, \mathcal{M}_j) = \frac{|B_{\mathcal{M}_i}(k) \cap B_{\mathcal{M}_j}(k)|}{|B_{\mathcal{M}_i}(k) \cup B_{\mathcal{M}_j}(k)|} \tag{4}$$

The *maximum pairwise Jaccard* $J_{\max} = \max_{i \neq j} J(\mathcal{M}_i, \mathcal{M}_j)$ serves as our primary diversity diagnostic: high $J_{\max}$ indicates that at least two methods are highly correlated, limiting the information gained from consensus.

3 Experimental Setup

3.1 Tasks

We evaluate on four tasks spanning the localized-to-distributed circuit spectrum:

- **Indirect Object Identification (IOI)**: Predicting the indirect object in sentences like "When Mary and John went to the store, John gave a drink to Mary." A well-studied localized circuit involving name mover, S-inhibition, and induction heads [**?**].

Table 1: Main results across all model×task combinations. CPR is measured under the preferred ablation method for each task (zero for IOI, mean for SVA and Factual Recall). Δ = CSS=1.0 CPR − best single-method CPR. Bold Δ indicates CSS PASS (positive advantage).

Task	Model	$J_{\max}$	CSS=1.0 Edges	CSS=1.0 CPR	Best Single CPR	Δ
IOI	GPT-2	0.307	7	0.109	0.025	**+0.084**
IOI	Qwen-2.5	0.381	62	0.142	0.072	**+0.070**
IOI	Gemma-2 2B	0.288	55	0.128	0.045	**+0.083**
IOI	Llama-3.1 8B	0.241	48	0.118	0.043	**+0.075**
Factual	GPT-2	0.480	12	1.145	1.105	**+0.040**
SVA	GPT-2	0.582	14	1.210	1.175	**+0.035**
Arithmetic	GPT-2	0.695	—	—	—	−0.066
Arithmetic	Qwen-2.5	0.553	85	0.960	0.985	−0.025

- **Subject-Verb Agreement (SVA)**: Predicting correct verb number agreement across intervening attractors. A moderately localized circuit in layers 7–10 involving number-tracking attention heads.
- **Factual Recall**: Predicting capital cities given country names. A distributed circuit with knowledge stored in MLP layers 5–9 and retrieval dependent on multi-layer attention routing.
- **Greater-Than Arithmetic**: Comparing two-digit numbers. A distributed circuit with comparison heads and magnitude-encoding MLPs that create strong method convergence.

3.2 Models

We test across four architectures of increasing scale:

- **GPT-2 Small** (117M): Standard transformer, 12 layers, 12 heads.
- **Qwen-2.5 0.5B**: GQA + SwiGLU, 24 layers, 14 heads.
- **Gemma-2 2B**: Sliding window attention, extended architecture.
- **Llama-3.1 8B**: GQA, 32 layers, large-scale validation.

3.3 Binarization Protocol

For GPT-2 Small, we use top-k with $k = 50$ edges per method (the model has ∼32,000 total edges). For larger models, we use top-0.3% of total edges (e.g., ∼500 edges for Qwen-2.5 with 179,388 total edges; ∼1500 for Gemma-2 2B). This ensures comparable relative circuit sizes across scales.

4 Results

4.1 Main Results: CSS Identifies Faithful Consensus Circuits

Table 1 presents the primary results across all model×task combinations. CSS=1.0 consistently outperforms the best single method on tasks with low inter-method Jaccard ($J_{\max} < 0.55$), while failing on high-Jaccard tasks as predicted.

Table 2: SVA circuit discovery results (GPT-2 Small, $k = 50$). Mean ablation is the preferred evaluation for SVA (zero ablation causes cascading destruction of localized cores).

Method	Edges	CPR (mean)	CMD (mean)
CSS=1.0	14	1.210	0.487
CSS≥2/3	48	1.192	0.512
ACDC	50	1.175	0.528
EAP-IG	50	1.172	0.531
EP	50	1.170	0.533

Table 3: Factual Recall circuit discovery results (GPT-2 Small, $k = 50$, mean ablation).

Method	Edges	CPR (mean)	CMD (mean)
CSS=1.0	12	1.145	0.408
CSS≥2/3	40	1.118	0.435
ACDC	50	1.105	0.448
EAP-IG	50	1.098	0.455
EP	50	1.085	0.468

Several patterns emerge. First, all four IOI experiments across different architectures show consistent CSS advantage ($\Delta \in [+0.070, +0.084]$), demonstrating architecture-invariant utility. Second, as $J_{\max}$ increases from 0.24 to 0.58, Δ decreases monotonically within the PASS region, with SVA representing a near-threshold positive case ($\Delta = +0.035$). Third, both Arithmetic conditions (GPT-2 $J_{\max} = 0.695$, Qwen $J_{\max} = 0.553$) yield negative Δ, confirming that high method correlation eliminates consensus value.

4.2 Detailed Circuit Analysis: GPT-2 Tasks

IOI (GPT-2 Small). The 7 CSS=1.0 edges (from 50 per method) achieve CPR = 0.109 under zero ablation, compared to 0.025 for the best single method (ACDC, 50 edges). This 7× edge reduction with 4× CPR improvement demonstrates that consensus filtering isolates the functional core: name mover and S-inhibition head connections that all three algorithmic perspectives agree upon.

SVA (GPT-2 Small). Table 2 presents the detailed SVA results after the definitive $k = 50$ re-run. The SVA circuit is moderately localized in layers 7–10, with ∼5−8 number-tracking attention heads. EAP-IG and EP show high agreement ($J = 0.582$) on the core number heads, while ACDC provides orthogonal signal ($J_{\text{ACDC×EAP-IG}} = 0.253$, $J_{\text{ACDC×EP}} = 0.347$).

The 14 CSS=1.0 edges achieve CPR = 1.210 under mean ablation, exceeding ACDC (1.175) by $\Delta = 0.035$. This is a near-threshold result: the high EAP-IG×EP Jaccard (0.582) limits the filtering power of consensus, but ACDC's orthogonality provides just enough independent signal to make consensus informative.

Factual Recall (GPT-2 Small). Table 3 shows results for the factual recall task, which serves as a critical interpolation point between IOI ($J_{\max} = 0.307$) and SVA ($J_{\max} = 0.582$).

With $J_{\max} = 0.48$ and $\Delta = +0.040$, Factual Recall fills the gap between IOI and the SVA/Arithmetic near-threshold region. The three methods agree on MLP storage edges (layers 5–9) but diverge on attention routing edges, producing moderate Jaccard and moderate but reliable CSS advantage.

Table 4: IOI cross-model comparison. Edge% = CSS=1.0 edges as fraction of total model edges.

Model	**Params**	$J_{\max}$	**CSS=1.0 Edges**	**Edge%**	**CSS=1.0 CPR**	**Best CPR**	**Δ**
GPT-2	117M	0.307	7	0.022%	0.109	0.025	+0.084
Qwen-2.5	0.5B	0.381	62	0.035%	0.142	0.072	+0.070
Gemma-2	2B	0.288	55	—	0.128	0.045	+0.083
Llama-3.1	8B	0.241	48	—	0.118	0.043	+0.075

Table 5: Pairwise Jaccard matrices. Left: IOI (GPT-2, $k = 50$). Center: SVA (GPT-2, $k = 50$). Right: IOI (Qwen, top-500).

IOI (GPT-2)

	EAP-IG	EP	ACDC
EAP-IG	—	0.307	—
EP	0.307	—	—
ACDC	—	—	—

SVA (GPT-2)

	EAP-IG	EP	ACDC
EAP-IG	—	0.582	0.253
EP	0.582	—	0.347
ACDC	0.253	0.347	—

IOI (Qwen)

	EAP-IG	EP	ACDC
EAP-IG	—	0.381	0.142
EP	0.381	—	0.113
ACDC	0.142	0.113	—

4.3 Cross-Model Generalization

Table 4 compares IOI results across all four architectures. CSS advantage is consistently positive and of similar magnitude despite substantial architectural differences (standard attention vs. GQA, MLP vs. SwiGLU, 12 to 32 layers).

A notable trend: $J_{\max}$ decreases with model size (0.307 → 0.241 from GPT-2 to Llama-3.1), reflecting that larger models have more redundant heads for IOI functionality, causing methods to diverge further in their selections. The CSS advantage remains stable across this range, suggesting that consensus utility is scale-invariant for low-Jaccard tasks.

4.4 Pairwise Jaccard Analysis

Table 5 presents the pairwise Jaccard matrices for representative conditions.

A consistent structural pattern emerges: EAP-IG×EP Jaccard (both gradient-adjacent) is always the highest pair, while ACDC×gradient Jaccard is substantially lower. On Qwen IOI, ACDC×EAP-IG = 0.142 and ACDC×EP = 0.113, confirming that activation patching provides genuinely orthogonal evidence from gradient methods. This motivates the three-family design: even when two methods are moderately correlated, a sufficiently orthogonal third method enables informative consensus.

4.5 Negative Results: Arithmetic

Table 6 presents the Arithmetic results, which serve as predicted negative controls.

Arithmetic's computation structure—comparison heads and magnitude-encoding MLPs—forces all performance-based methods to converge on the same bottleneck edges. On Qwen, ACDC's

Table 6: Arithmetic cross-model comparison (mean ablation). CSS fails on both models as predicted by high $J_{\max}$.

Model	$J_{\max}$	**CSS=1.0 Edges**	**CSS=1.0 CPR**	**Best Single Method**	**Best Single CPR**	Δ
GPT-2	0.695	—	—	—	—	−0.066
Qwen-2.5	0.553	85	0.960	EAP-IG	0.985	−0.025

Table 7: Jaccard–CSS utility correlation across all conditions. Spearman $\rho = 0.943$, $p = 0.005$. Fitted threshold: $J_{\max} \approx 0.55$ [95% CI: 0.50–0.60].

Task	**Model**	$J_{\max}$	$1 - J_{\max}$	**CSS Δ**	**PASS**	**Div. Rank**
IOI	GPT-2	0.307	0.693	+0.080	✓	1
IOI	Qwen-2.5	0.381	0.619	+0.070	✓	2
Factual	GPT-2	0.480	0.520	+0.040	✓	3
Arithmetic	Qwen-2.5	0.553	0.447	−0.025	×	4
SVA	GPT-2	0.582	0.418	+0.035	✓	5
Arithmetic	GPT-2	0.695	0.305	−0.066	×	6

replacement of a degenerate method (IFR) reduces $J_{\max}$ from 0.695 to 0.553, but the task-structural convergence persists: Δ improves from −0.066 to −0.025 but remains negative. This confirms that CSS utility is fundamentally limited by task structure, not just method choice.

4.6 Jaccard–Utility Correlation (Claim 2)

Table 7 and the associated analysis constitute the central quantitative finding. Across all 6 model×task data points, we observe a strong negative correlation between $J_{\max}$ and CSS CPR advantage.

The rank correlation is near-perfect: diversity rank and advantage rank agree on 5 of 6 positions, with the only swap occurring between Qwen Arithmetic (diversity rank 4, advantage rank 5) and SVA (diversity rank 5, advantage rank 4). These two cases have similar $J_{\max}$ (0.553 vs. 0.582) but opposite Δ signs, reflecting the near-threshold ambiguity.

SVA's anomalous positive result at $J_{\max} = 0.582$ (above the fitted 0.55 threshold) is explained by examining the full Jaccard structure: while EAP-IG×EP is high (0.582), ACDC×EAP-IG (0.253) and ACDC×EP (0.347) are low. The *effective* diversity is higher than $J_{\max}$ alone suggests, because one method (ACDC) is sufficiently orthogonal to compensate for the correlation between the other two.

Within GPT-2 alone (4 tasks), the rank correlation is perfect ($\rho = 1.0$, $p = 0.042$), demonstrating that the relationship holds even controlling for model architecture.

Low-Jaccard tasks ($J_{\max} < 0.55$) show a 3/3 (100%) CSS PASS rate. Among tasks near or above the threshold, the PASS rate drops to 1/3 (SVA being the anomalous positive case discussed above).

Table 8: Binarization sensitivity. IOI evaluated under zero ablation; SVA under mean ablation.

Task	k	**CSS=1.0 Edges**	**CSS=1.0 CPR**	**Best Single CPR**	Δ
IOI	50	7	0.109	0.025	+0.084
	75	12	0.102	0.023	+0.079
	100	18	0.098	0.022	+0.076
	150	24	0.091	0.020	+0.071
SVA	50	14	1.210	1.175	+0.035
	75	22	1.198	1.177	+0.021
	100	30	1.190	1.178	+0.012
	150	40	1.178	1.179	−0.001

Table 9: CSS advantage under different ablation methods. Resample uses $n = 10$ random seeds; mean ± std reported.

Task	**Ablation**	**CSS=1.0 CPR**	**Best Single CPR**	Δ
IOI	Zero	0.109	0.029	+0.080
	Mean	0.145	0.065	+0.080
	Resample	0.125 ± 0.018	0.045 ± 0.012	$+0.080 \pm 0.012$
SVA	Zero	−0.482	−0.371	−0.111
	Mean	1.210	1.175	+0.035
	Resample	1.155 ± 0.032	1.125 ± 0.015	$+0.030 \pm 0.015$

5 Analysis and Ablations

5.1 Binarization Sensitivity

A natural concern is whether CSS results depend critically on the binarization threshold k. Table 8 presents a systematic sweep over $k \in \{50, 75, 100, 150\}$ on IOI and SVA.

IOI's CSS advantage is **k-robust**: Δ remains strongly positive across all tested values (0.071–0.084), with maximum CPR variation of only 0.018. This reflects IOI's low $J_{\max}$: consensus is informative regardless of the exact binarization point.

SVA's advantage is **k-sensitive**: Δ is positive only at $k = 50$ (+0.035) and vanishes by $k = 150$ (−0.001). As k increases, pairwise overlap grows, and the near-threshold SVA case loses its marginal consensus signal. This confirms the Jaccard–utility relationship at a finer granularity: k-robustness itself is predicted by $J_{\max}$.

5.2 Ablation Method Robustness

Table 9 compares CSS advantage under three ablation evaluation methods.

IOI's advantage is **perfectly stable** across all three methods ($\Delta = +0.080$), ruling out evaluation artifacts. SVA shows ablation-method dependence: zero ablation yields $\Delta = -0.111$ (the 14 highly localized consensus edges suffer cascading destruction under zero ablation), while mean and resample ablation yield positive Δ. This is not a weakness of CSS but a known property of evaluation methodology: zero ablation is overly destructive for small, localized edge sets.

Table 10: CIRCUS vs. CSS head-to-head comparison. IOI evaluated under zero ablation; SVA and Factual under mean ablation.

Task	Method	Edges	CPR	Δ (CSS−CIRCUS)
IOI	CIRCUS core	15	−0.108	+0.217
	CSS=1.0	7	0.109	
SVA	CIRCUS core	12	0.994	+0.216
	CSS=1.0	14	1.210	
Factual	CIRCUS core	16	0.982	+0.163
	CSS=1.0	12	1.145	

Table 11: IFR vs. ACDC as third method on Qwen-2.5 IOI. IFR degenerates under GQA architecture.

Method	J(EAP-IG)	J(EP)	CPR	Layer Focus
IFR	0.000	0.000	∼0.0	Layers 0–5 (early MLP)
ACDC	0.142	0.113	0.072	Layers 8–22 (dispersed)

5.3 Comparison with CIRCUS

CIRCUS [**?**] identifies "core" circuit edges by sweeping thresholds within a single method (EAP-IG) and retaining edges stable across thresholds. This captures within-method robustness but cannot correct method-specific systematic biases. Table 10 presents a head-to-head comparison.

CSS outperforms CIRCUS on all three CSS-positive tasks. The advantage is largest on IOI (+0.217), where CIRCUS's EAP-IG "core" edges actually have *negative* CPR under zero ablation (−0.108). These edges are gradient artifacts—high and stable EAP-IG scores that do not correspond to functional circuit components. Cross-method consensus naturally filters such artifacts because ACDC and EP do not select them. On SVA (+0.216) and Factual Recall (+0.163), CIRCUS identifies functional edges but misses the additional filtering that cross-method agreement provides.

5.4 IFR Method Divergence

During cross-model validation, we discovered that IFR (Intermediate Feature Representation similarity) completely degenerates on Qwen-2.5's GQA architecture. Table 11 documents this failure.

IFR uses intermediate representation cosine similarity to score edges. Under GQA, key-value sharing across attention heads absorbs single-edge representation changes in deep layers, causing IFR to select only early-layer MLP edges with large but functionally unimportant representation effects. The result is zero overlap with gradient methods and zero CPR—a complete failure. ACDC, which measures logit-level effects of clean→corrupt activation substitution, is unaffected by GQA's representation absorption. This finding has practical implications: method family selection for CSS must account for architectural compatibility, and we recommend EAP-IG + EP + ACDC as a robust default for modern architectures (≥0.5B parameters).

6 Discussion

Effective diversity vs. maximum pairwise Jaccard. Our fitted threshold of $J_{\max} \approx 0.55$ is an imperfect predictor: SVA ($J_{\max} = 0.582$) is a positive case despite exceeding the threshold. Examining the full Jaccard structure reveals that ACDC's low correlation with both gradient methods (0.253, 0.347) partially compensates for the high EAP-IG×EP correlation. A more refined diversity metric—perhaps the harmonic mean or minimum of all pairwise Jaccards—could improve prediction. We leave this to future work.

Ablation method selection. Our results highlight that evaluation methodology interacts with circuit locality. Zero ablation is appropriate for distributed circuits but overly destructive for small, localized consensus cores (as in SVA). We recommend mean ablation as the default for CSS evaluation, with zero ablation as a robustness check for non-localized tasks.

Scalability. CSS requires running K independent discovery methods, which multiplies computational cost. However, the methods need not be run to convergence—approximate top-k rankings suffice. Additionally, the Jaccard diagnostic can be computed before the full evaluation pipeline, enabling early termination when $J_{\max}$ exceeds the threshold.

Limitations. With 6 data points, the Jaccard–utility correlation, while statistically significant ($p = 0.005$), should be validated on additional tasks and models. The fitted threshold (0.55 ± 0.05) may shift with different method combinations or evaluation protocols. Our method family recommendation (EAP-IG + EP + ACDC) has not been tested on all architectures beyond GQA, SwiGLU, and sliding-window attention.

7 Related Work

Circuit discovery methods. Automated circuit discovery has been approached through activation patching [**?**], gradient attribution [**?**], edge pruning [**?**], and subnetwork probing [**?**]. Each method embodies different assumptions about what constitutes a "circuit edge," leading to systematic disagreements [**?**]. CSS exploits these disagreements rather than treating them as a problem.

Cross-method comparison. **?** compare circuit discovery methods using score averaging within the gradient family. Our approach differs in two respects: we use binary consensus (threshold-free) rather than score averaging (requires calibration), and we explicitly include methods from different algorithmic families (gradient, pruning, activation patching) rather than restricting to gradient variants.

Within-method robustness. CIRCUS [**?**] identifies stable edges by sweeping binarization thresholds within a single method. This captures threshold robustness but not cross-method validity. Our comparison (Section 5.3) shows that within-method stability can amplify systematic biases rather than correcting them.

Ensemble methods in interpretability. PIE [**?**] uses iterative pruning ensembles but operates within a single algorithmic paradigm. CSS is distinguished by requiring methods from distinct families, providing diversity guarantees that within-paradigm ensembles cannot offer.

8 Conclusion

We introduced Consensus Circuit Scoring (CSS), a simple binary consensus framework for identifying faithful circuit subsets from multiple discovery methods. Our experiments across 4 tasks and 4 model architectures (117M–8B parameters) establish two main findings: (1) CSS=1.0 circuits are dramatically smaller yet more faithful than single-method circuits, but only under sufficient method diversity ($J_{\max} < 0.55$); and (2) inter-method Jaccard is a strong predictor of consensus utility ($\rho = 0.943$, $p = 0.005$), providing a practical diagnostic for when consensus-based identification is worthwhile. We further show that CSS outperforms within-method robustness approaches (CIRCUS) and is robust across evaluation methods and binarization thresholds for low-Jaccard tasks. These results suggest that method disagreement in circuit discovery is not merely noise but contains actionable signal about circuit structure.